

```
49 // Continues from Figure 5.3.
50 // Define some initial values for the optimizer.
51 Values initialValues = ...
52
53 // Set up and run the optimizer.
54 GaussNewtonOptimizer opt(graphInterp, initialValues);
55 Values resultBorder = opt.optimize(); // only contains the border states.
56
57 // Fill in the interpolated states using after-solve interpolation.
58 // Construct an empty covariance map to hold the output
59 auto covarianceMap = std::make_shared<CovarianceMap>();
60
61 // Interpolate
62 Values resultAll = updateInterpValues<Pose3>(graphInterp, resultBorder,
63 borderStates, interpStates, Q_se3, covarianceMap);
64
65 // The means and covariances for all states can be extracted from
66 // resultsAll and covarianceMap, respectively.
```